

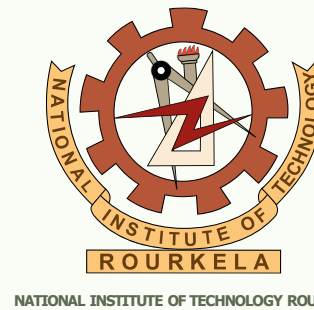
Deep Eutectic Solvent Electrolytes for Sustainable Lithium–Sulphur Batteries

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01 INTRODUCTION

The growing demand for renewable energy and electric vehicles requires safe, high-energy, and sustainable batteries. Lithium–Sulphur (Li–S) batteries offer a high theoretical energy density (2600 Wh kg⁻¹) and specific capacity (1675 mAh g⁻¹), but their practical application is limited by the polysulfide shuttle effect and flammable organic electrolytes.

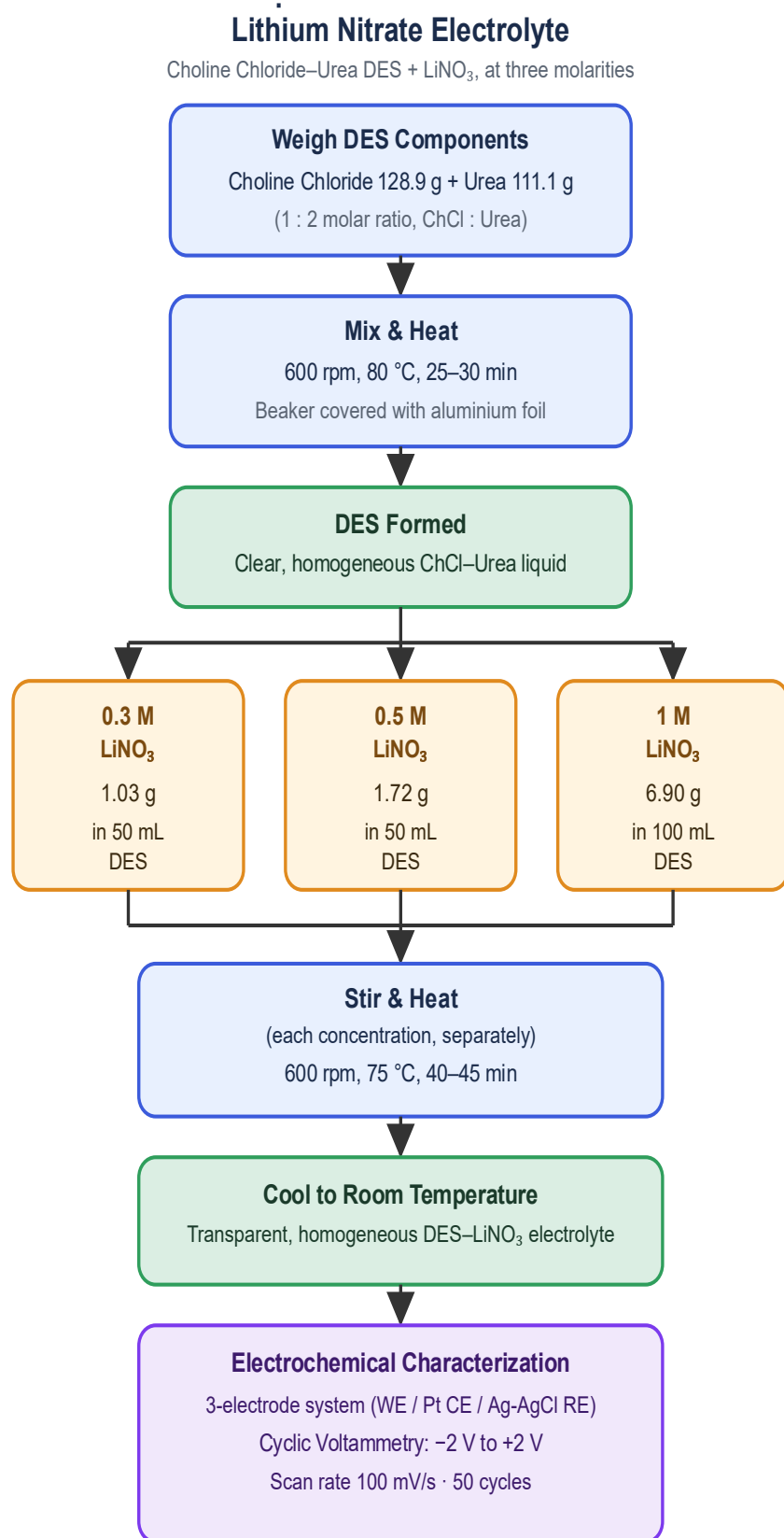
Deep Eutectic Solvents (DESs) are promising green electrolytes due to their low toxicity, low volatility, and high thermal stability. This study investigates DES electrolytes using cyclic voltammetry (CV) to identify an optimum electrolyte for future Li–S battery applications.

02 OBJECTIVES

- ✓ Prepare a Choline Chloride–Urea Deep Eutectic Solvent (DES).
- ✓ Develop DES electrolytes with different LiNO₃ concentrations (0.3 M, 0.5 M and 1 M).
- ✓ Investigate electrochemical behaviour using Cyclic Voltammetry (CV).
- ✓ Identify the optimum LiNO₃ concentration for developing a safe and sustainable electrolyte for future Li–S batteries.

03 METHODOLOGY

DES electrolytes with varying LiNO₃ concentrations (0.3–1.0 M) were synthesized and characterized using cyclic voltammetry (CV) in a three-electrode cell. The experimental workflow is summarized below



Development of an Alternative Safe Electrolyte for Electrochemical Storage Systems

Figure 1. Schematic of the proposed methodology.

04 RESULTS

Cyclic voltammetry (CV) was used to evaluate the electrochemical performance of DES electrolytes. Variations in the current response with LiNO₃ concentration were used to compare ion transport and electrochemical activity.

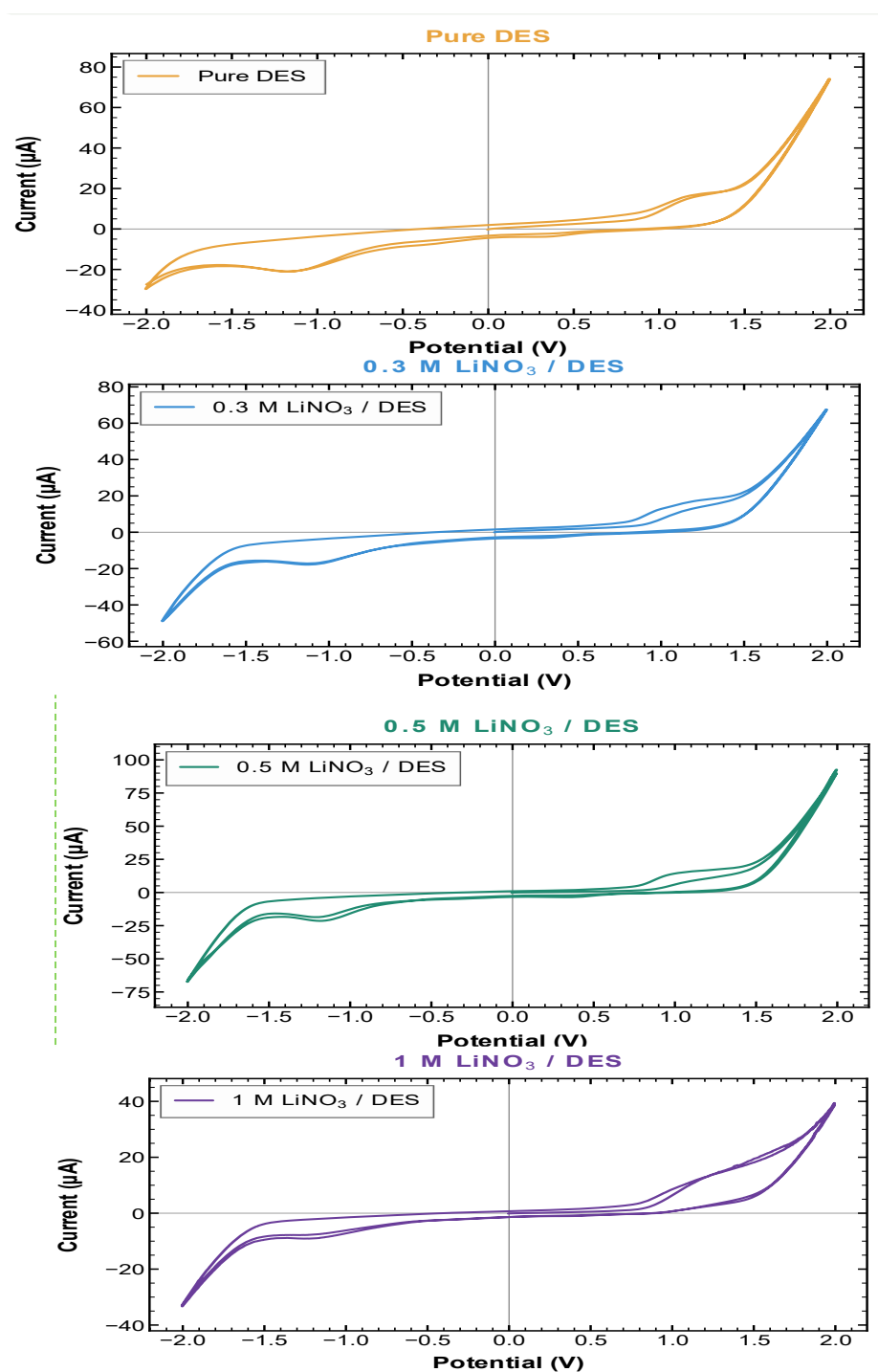
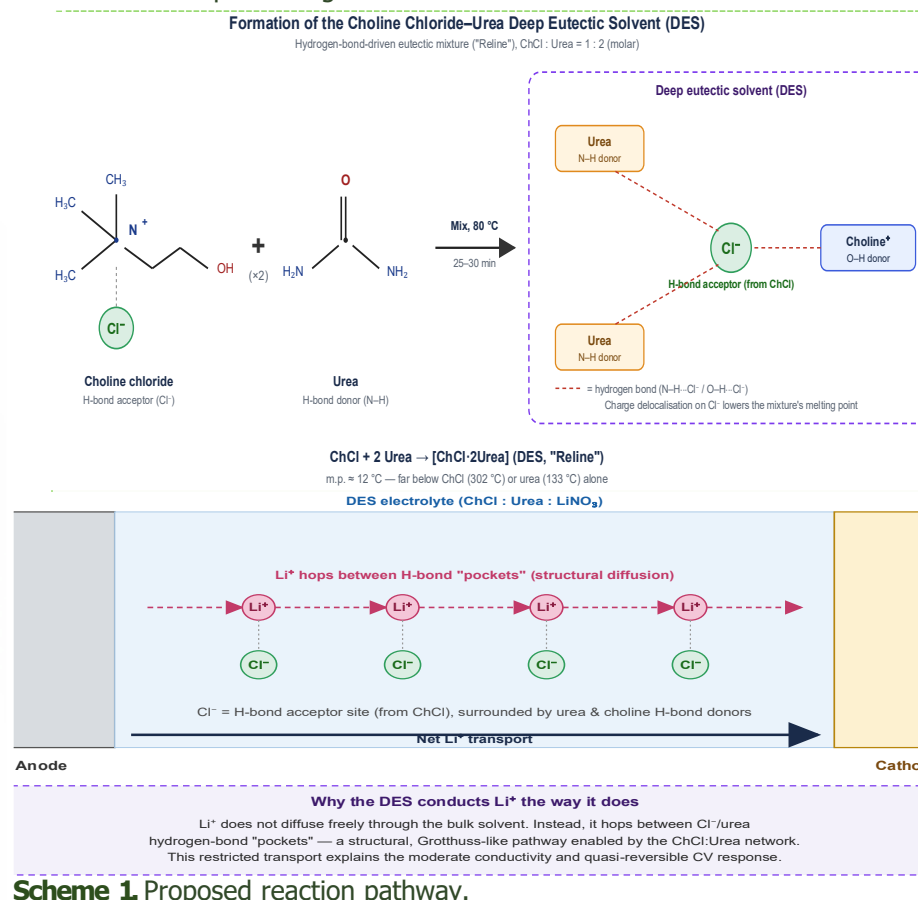


Figure 2. CV Plot for all Composition

PARAMETER	BASELINE	THIS WORK	Δ %
Peak Current (μA)	74.2	92.63	+24.8
Peak Potential (V)	+1.99	+1.99	+0.0
Current Density (mA/cm ²)	1.05	1.31	+24.5
CV Loop Area (μA · V)	36.22	33.39	–7.8

Table 1. Comparison against Pure DES and 0.5 M LiNO₃.



Scheme 1. Proposed reaction pathway.

05 DISCUSSION

The 0.5 M LiNO₃ electrolyte exhibited the most favourable redox response, indicating an optimum balance between Li⁺ availability and ion transport. However, DES electrolytes still face challenges such as relatively high viscosity and ion pairing, which can restrict Li⁺ mobility and limit electrochemical performance. Further optimization of the electrolyte composition is therefore required.

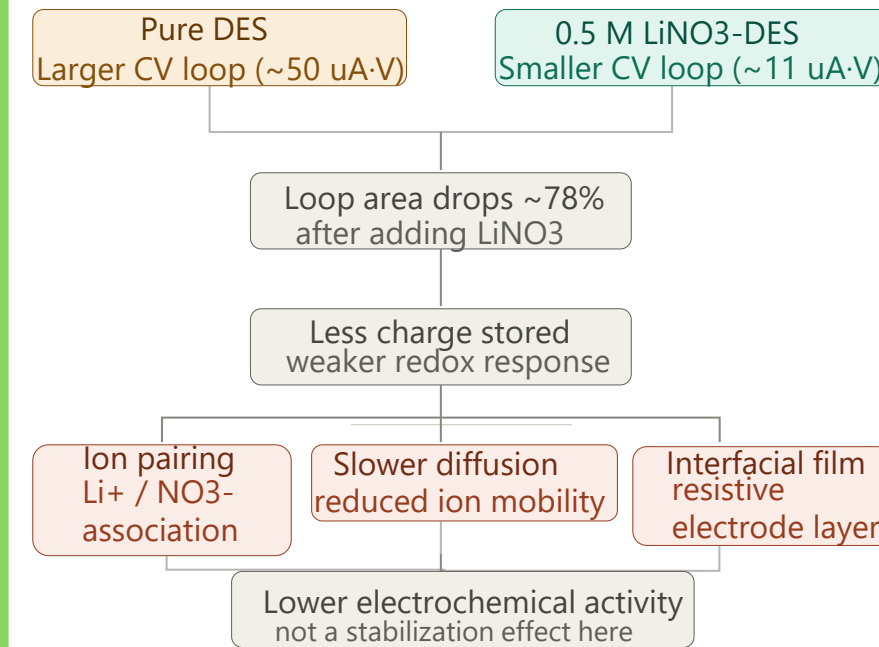
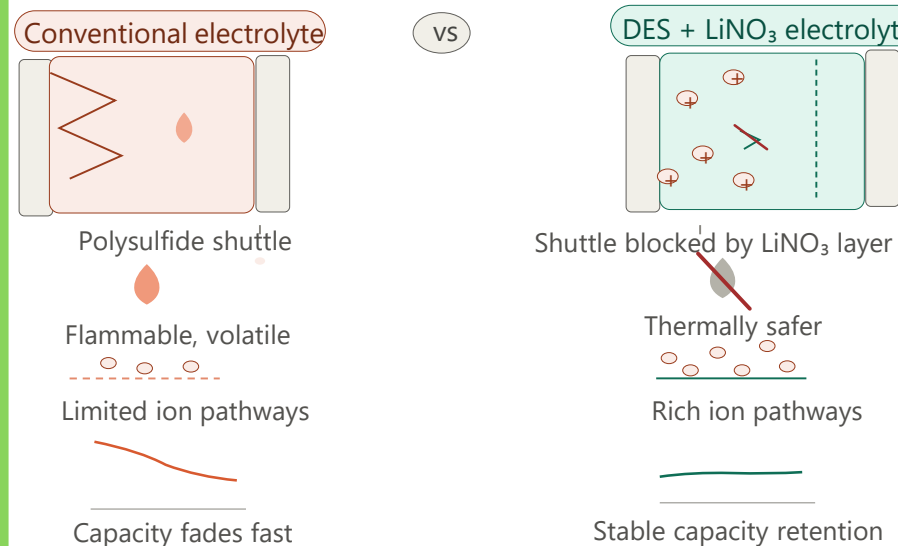


Figure 3. Supporting evidence for the discussion.

06 CONCLUSIONS

- 0.5 M LiNO₃ showed the highest electrochemical activity among all.
- DES offer a safer and greener alternative to conventional electrolytes.
- The developed electrolyte shows promise for future lithium–sulphur battery applications.

Deep Eutectic Solvent electrolytes represent a promising route toward safer, environmentally benign, and next-generation Li–S batteries.

07 FUTURE WORK

Future work will optimize the DES composition using alternative lithium salts (LiTFSI and LiFSI) and suitable additives to improve ionic conductivity and reduce viscosity. The optimized electrolyte will be validated through EIS, long-term cycling, and Li–S battery testing for sustainable energy storage applications.

08 REFERENCES

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09 ACKNOWLEDGEMENTS

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